MACHINE LEARNING CASE STUDY · SUMMARY

Predicting 30-Day Patient Readmission

A two-page summary of the student brief: scenario, dataset, tasks, marking and submission

100 marks

Formative assessment

LEVEL	MODULE	ASSESSMENT	DATASET	SUBMIT
Postgraduate	Programming (Python)	Formative · 100 marks	3,000 admissions · 10 fields	Report (PDF) + notebook

The scenario

A healthcare startup wants a machine learning model that predicts which patients will be readmitted to hospital within 30 days of discharge, so that hospitals can target follow-up care, cut avoidable readmissions and improve patient outcomes. It has supplied historical patient data and expects the model to be built in Python with its data science libraries, and to be **understandable** and **honestly evaluated**, not only accurate.

MAIN QUESTION

Develop a predictive model using Python to assess the likelihood of patient readmission within 30 days of discharge. Use Python's data science libraries to process and analyse the data, build a predictive model and evaluate its effectiveness.

What you hand in

- ✓ A **report** (PDF) in the structure on page 2, written for a technical lead and a hospital manager
- ✓ A **notebook** (.ipynb or .py) that runs top to bottom on the supplied CSV, outputs visible, sections matching the five tasks
- ✓ A **declaration** of sources, collaborators and any AI tools used, and for what

The dataset

One row per hospital admission. File: `Expanded_Patient_Readmission_Data_1.csv`. Column names contain spaces and capitals; rename them in code if you wish, and say so.

FIELD	DESCRIPTION	VALUES
Patient ID	Identifier for the admission record	1–3000
Age	Age at admission (years)	18–89
Gender	As recorded	Female / Male
Admission Type	Planned or unplanned	Elective / Emergency
Length of Stay	Days in hospital	1–29
Number of Diagnoses	Recorded during the stay	1–9
Blood Pressure	Reading at admission (mmHg)	60–179
Blood Sugar Levels	Reading at admission (mg/dL)	51–199
Previous Admissions	Earlier admissions on record	0–4
Readmission	**Target** · readmitted within 30 days	Yes / No

The file is supplied as exported. Verifying its quality (types, ranges, duplicates, missing values, label consistency) is part of Task 1, whatever you find. Treat it as real patient data: do not share or link it.

Three rules that apply to every task

1. **Justify every decision**
   Why this imputation, this encoding, this threshold? A sentence of reasoning beats a page of code.
2. **Never let the test set leak**
   Split before fitting any scaler, imputer or model; keep preprocessing inside a Pipeline.
3. **Interpret, do not just display**
   Every chart, table and metric is followed by what it means for the hospital.

You are marked on method, evidence and reasoning, not on reaching a particular score.

A carefully built model with weak predictive power, reported honestly with a discussion of why, is a strong submission. An impressive score with no explanation of how it was validated is a weak one.

The five tasks at a glance

TASK	WHAT TO DO	SHOW IN YOUR REPORT
1 **Data preprocessing and cleanup** 20 marks	• Inspect and audit: types, missing values, duplicates, labels • Treat missing, invalid and outlying values with a stated rule • Encode Gender, Admission Type and the target; scale numerics • Stratified train/test split *before* fitting imputer or scaler	✓ Audit and before/after summary ✓ Cleaning rules with reasons ✓ Encoding, scaling and split choices
2 **Exploratory data analysis** 20 marks	• Target balance and what it means for your metrics • Distributions of every numeric and categorical variable • Rates by group; box plots by outcome; correlation heatmap • Interpret every figure; list the insights you carry forward	✓ Five or more labelled figures ✓ Descriptive statistics table ✓ A sentence under every figure
3 **Feature selection** 20 marks	• Chi-square for categories; t-test / point-biserial for numbers • Mutual information; forest, L1 or permutation importance • Drop redundant and inappropriate features (identifiers) • Keep/drop decision with evidence for every feature	✓ Results table per feature ✓ Feature importance chart ✓ Keep/drop table; final features
4 **Model building** 20 marks	• Baseline; logistic regression, decision tree, random forest • Class imbalance: class weights, in-fold resampling, threshold • Stratified k-fold cross-validation on ROC-AUC • Tune two or more models (`GridSearchCV`); select with reasons	✓ Candidates: CV mean ± SD ✓ Search space and best parameters ✓ Reasoned model selection
5 **Model evaluation and testing** 10 marks	• Test set, once: accuracy, precision, recall, F1, ROC-AUC • Confusion matrix, ROC curve and precision–recall curve • Justify a decision threshold; check the train vs test gap • Overfitting mitigation; limitations and ethics	✓ Metrics, confusion matrix, curves ✓ Overfitting and threshold ✓ Limitations paragraph

Report structure and marks

2	2	2	20	20	20	20	10	2	2
Title page	Table of contents	Introduction	Data preprocessing	Exploratory data analysis	Feature selection	Model development	Model evaluation	Conclusion	List of references

Use these headings in this order. Caption every figure and table; long code in an appendix; Harvard references; 3,000–5,000 words excluding code and appendices.

Suggested timeline

Week 1 Tasks 1–2: data audited and cleaned; EDA charts interpreted; insights listed.

Week 2 Tasks 3–4: tests and importances; feature decision table; models cross-validated and tuned.

Week 3 Task 5 and write-up: test-set evaluation, threshold and overfitting; report drafted, referenced, proofread.

Submission

- `Surname_Initials_Readmission_Report.pdf` and `Surname_Initials_Readmission_Notebook.ipynb`, through the platform, by the deadline given by your facilitator
- Individual work: the code, analysis and writing must be your own, with any AI assistance declared

Before you submit

- ☐ The data was audited before it was changed, and every change has a reason
- ☐ The train/test split happens before any scaler, imputer or model is fitted
- ☐ Every figure has a title, labelled axes with units, a number and an interpretation
- ☐ The test set was used once, at the end; all five metrics plus curves are reported
- ☐ The conclusion answers the main question and gives the hospital a recommendation
- ☐ The notebook runs end to end and its numbers match the report